Machine Learning

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Machine Learning

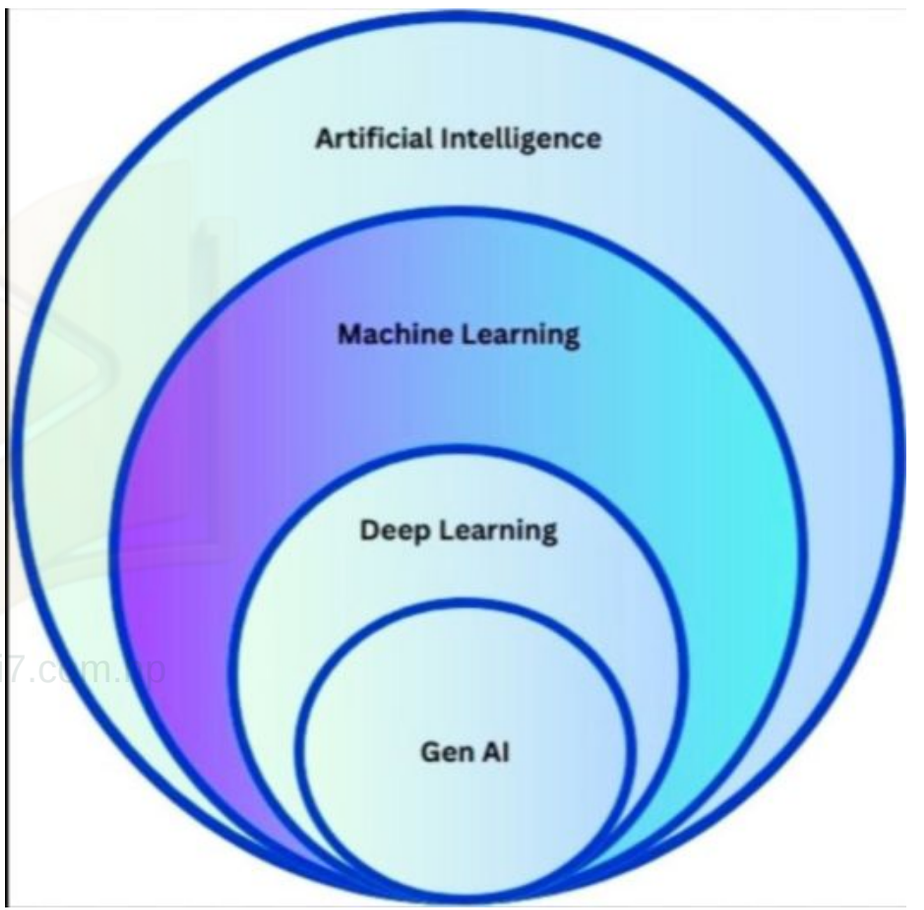
- What is learning and why needed?
- Why we want agent to learn? | we just improve the ploblem that is there in model?
  - Designer can't anticipate all possible situation, that agent might come across. e.g.: maze
  - Designer can't anticipate all changes over time. Eg : stock market
  - Human programmer don’t have idea to solve problem by themselves. Eg: face detection

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Machine Learning

- Field of Artificial Intelligence (AI) that enables systems to learn and improve from experience without being explicitly programmed.
- Focus on developing algorithms that can process data, identify patterns, and make decisions.
- Key characteristics
  - Data Driven Approach
  - Adaptive and Scalable
  - Predictive Modeling



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Evolution of Machine Learning

1. 1950s: Concept of learning machines introduced (Alan Turing’s "Learning Machine").
2. 1960s-70s: Symbolic AI and simple ML algorithms developed (e.g., Perceptron).
3. 1980s-90s: Neural networks and support vector machines (SVMs) emerged.
4. 2000s: Rise of deep learning and big data.
5. 2010s-Present: Advanced techniques like reinforcement learning, GANs, and real-world applications.

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Learning by Analogy

- Learning by drawing parallels between a new situation and previously encountered situations.
- Solving a problem in one domain by applying the solution from a different domain.
- Involves abstracting details from a particular set of problems and resolving structural similarities between previously distinct problems

Applications:

- Case-based reasoning systems
- Natural language processing (e.g., metaphor understanding)

Advantages:

- Intuitive and efficient for human-like learning.

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Explanation-Based Learning (EBL)

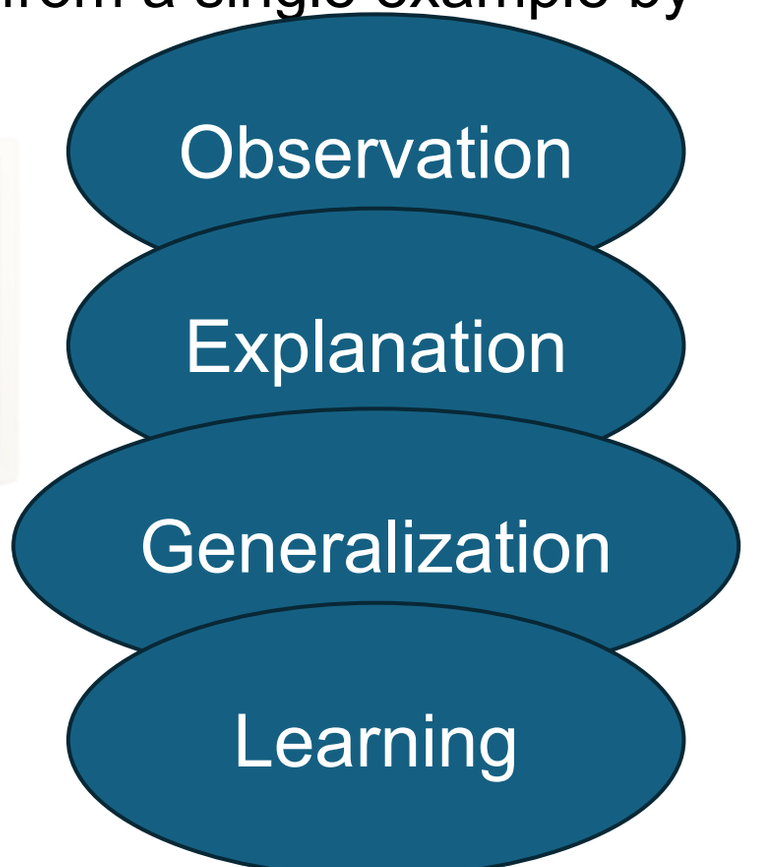
A learning technique where a system generalizes from a single example by understanding the underlying explanation.

Steps in EBL:

1. Analyze a specific example.
2. Extract the generalized rule or explanation.
3. Apply the rule to similar problems.

Applications:

- Knowledge-based systems.
- Robotics and automated planning.



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Feedback Based

- Supervised Learning
- Unsupervised Learning
- Reinforced Learning

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Supervised Learning

Learning from labeled data, where the algorithm maps inputs to known outputs.

Key Characteristics:

- - Requires a labeled dataset.
- - Goal: Minimize the error between predictions and actual outputs.

Applications:

- - Email Spam Classification
- - Stock Price Prediction
- - Image Recognition

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Two approaches of Supervised Learning

1. Classification: A classification problem is when the output variable is a category, such as “Red” or “blue” , “disease” or “no disease”.

    Example: Classifying emails as spam or not spam.

    Algorithms: Logistic Regression, Decision Trees, SVM, Random Forests, Naive Baye.

2. Regression: A regression problem is when the output variable is a real value, such as “dollars” or “weight”.

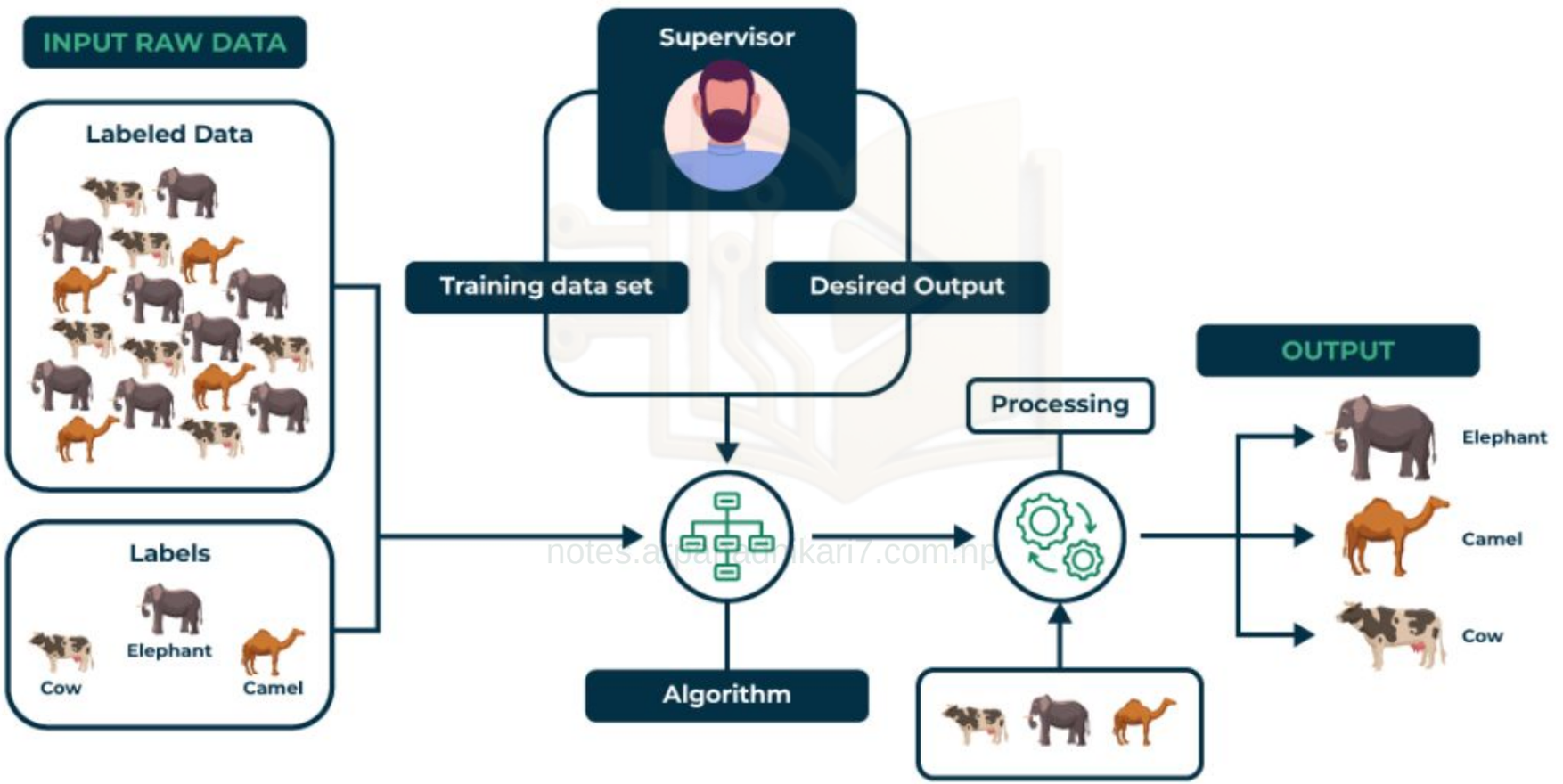
    Example: Predicting house prices.

    Algorithms: Linear Regression, Ridge Regression, Support Vector Machine Regression, Decision Tree Regression, Random Forest Regression.

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Supervised Learning



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Types of Supervised Learning Algorithms

- Classification
- Classification is a technique with which we can **affiliate** data under certain labels. We can train algorithms and models to classify data under different classes.
- For example, if you have data of fruits consisting of apples and oranges, your model will learn from the data about the specifications of the fruits.
- The next time you enter new data, the model will compare your data and then classify it under either class.

    Binomial Classification

    Multi-Class Classification
- Regression
- Regression helps us to understand the relationship between various data points and helps us to find hidden patterns among the data.
- We have both dependent and independent variables in this case which can help us to understand the relation with the help of **graphical** representations.

    Linear Regression
    Logistics Regression

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K nearest Neighbor (KNN)

- A non-parametric algorithm that classifies or predicts a data point based on the majority class (or average value) of its k-nearest neighbors.
- Working:
  - Choose the number of neighbors (‘k’).
  - Calculate distances between the data point and all other points.
  - Identify the k-nearest neighbors.
  - For classification: Assign the majority class.
  - For regression: Compute the average of neighbors.
- Distance Matrices
  - Euclidean
  - Manhattan
  - Minkowski

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Application of Supervised Learning

- prediction of stock markets
- Speech Recognition
- Object detection

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Unsupervised Learning

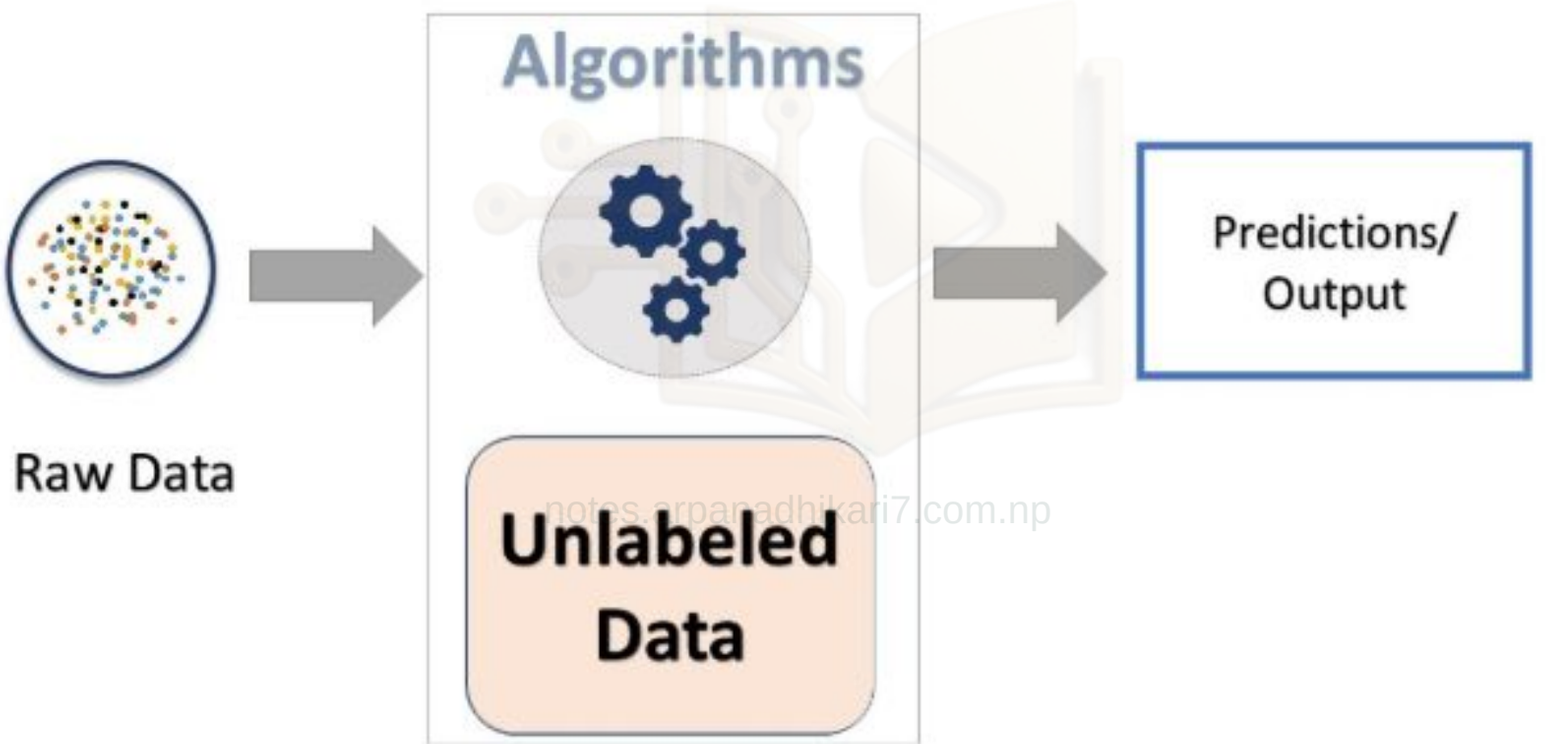
Learning from unlabeled data, identifying patterns and structures.

- Key Characteristics:
- - No labeled outputs.
- - Goal: Discover hidden structures or features in data.
- Applications:
- - Customer Segmentation
- - Fraud Detection
- - Dimensionality Reduction

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Unsupervised Learning



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Types of Unsupervised learning

1. Clustering:
   - Example: Grouping customers by purchasing behavior.
   - Algorithms: K-Means, Hierarchical Clustering.
2. Dimensionality Reduction:
   - Example: Reducing the number of features in a dataset.
   - Algorithms: PCA, t-SNE.

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Clustering

- An unsupervised learning task that groups data points into clusters based on similarity.
- Types of Clustering:
  - Centroid-based: K-means
  - Density-based: DBSCAN
  - Hierarchical: Agglomerative Clustering
- Applications:
  - Customer segmentation
  - Document categorization

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K Means Clustering

- A centroid-based clustering algorithm that divides data into k clusters by minimizing within-cluster variance.
- Algorithm Steps:
  - Choose the number of clusters (‘k’).
  - Initialize ‘k’ centroids randomly.
  - Assign each data point to the nearest centroid.
  - Update centroids by computing the mean of assigned points.
  - Repeat steps 3 and 4 until convergence.
- Use Cases:
  - Market segmentation
  - Image compression

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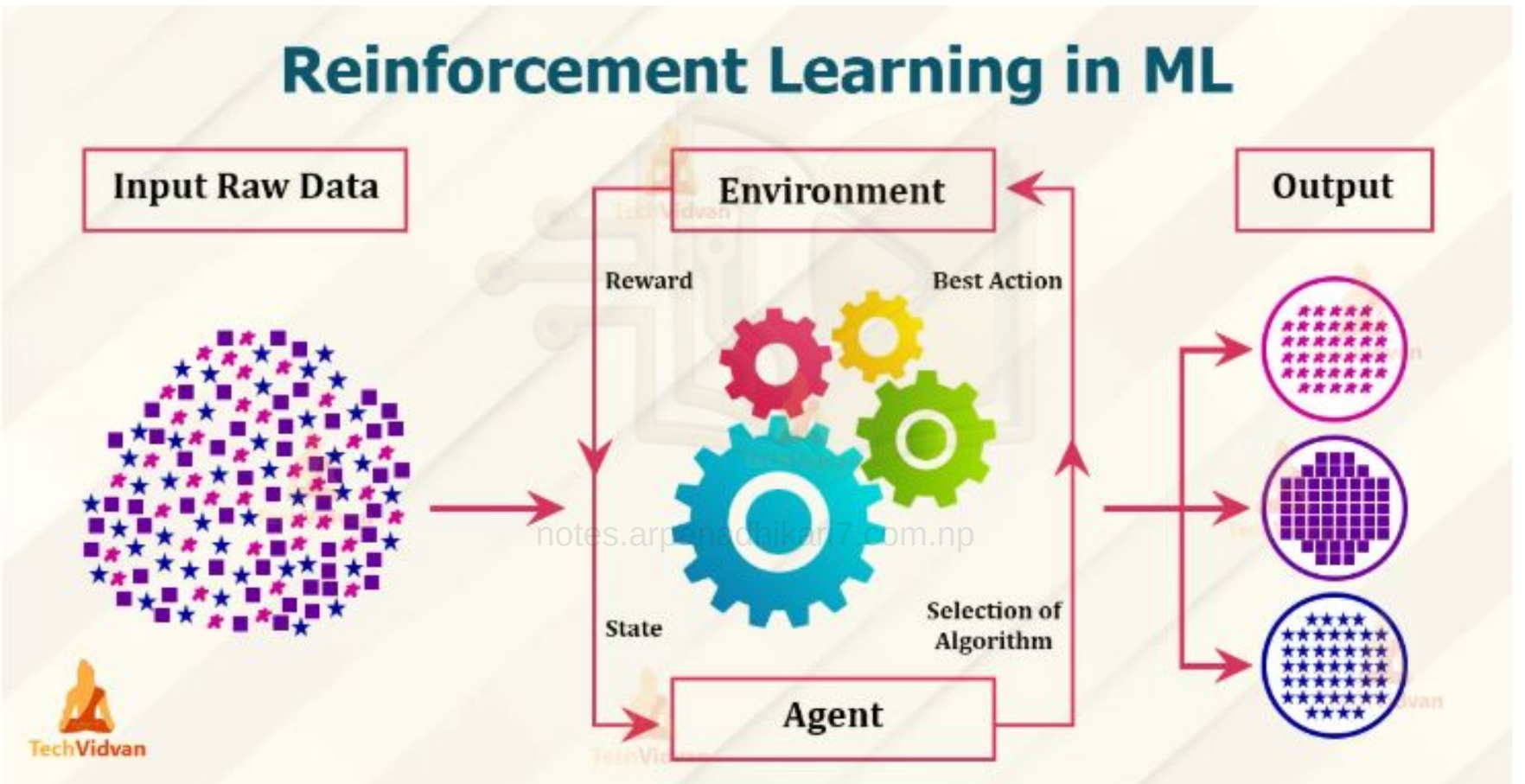
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Reinforcement Learning

- Definition: Learning by interacting with an environment to maximize cumulative rewards.
- Key Characteristics:
- - Involves agents, actions, states, and rewards.
- - Goal: Learn a policy for optimal decision-making.
- Applications:
- - Game Playing (e.g., Chess, Go)
- - Autonomous Vehicles
- - Robotics

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Types of Reinforcement Learning

- 1. Positive Reinforcement:
- - Reward for correct actions.
- - Example: Training a robot to walk.
- 2. Negative Reinforcement:
- - Penalty for incorrect actions.
- - Example: Learning to avoid obstacles in a maze.

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Comparision

- | Feature | Supervised Learning | Unsupervised Learning | Reinforcement Learning |
- |---|---|---|---|
- | Data Type | Labeled | Unlabeled | Interactive |
- | Goal | Predict output | Find patterns | Maximize rewards |
- | Applications | Image recognition, Spam segmentation | Customer | Game playing, Robotics |
- | Algorithms Examples | SVM, Decision Trees | K-Means, PCA | Q-Learning, Deep Q-Networks |

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